

**Cost-Effectiveness of Intismeran Autogene Plus
Pembrolizumab in Resected Stage IIIB–IV
Melanoma: An Early Economic Evaluation**

Blinded for peer review

Abstract

Background: In August 2026, Merck and Moderna reported positive Phase 3 INTerpath-001 results for intismeran autogene (mRNA-4157/V940) plus pembrolizumab in resected stage IIIB–IV melanoma. Its cost-effectiveness remains unknown.

Objective: To estimate the cost-effectiveness of intismeran plus pembrolizumab versus pembrolizumab alone from a US payer perspective.

Methods: A four-state partitioned survival model was developed with a 40-year horizon and 3% annual discounting. Quantitative efficacy inputs were derived from the Phase 2b KEYNOTE-942 trial (n=157), with exploratory OS; Phase 3 INTerpath-001 results were used as clinical context because numerical effect estimates were not yet available. Health-state utilities were drawn from published EQ-5D literature. Probabilistic, deterministic, and scenario sensitivity analyses were conducted.

Results: In the base case, intismeran plus pembrolizumab yielded 11.46 QALYs at a cost of \$881,300 versus 7.41 QALYs at \$719,353 for pembrolizumab alone. The ICER was \$39,976/QALY (incremental cost \$161,947). Under the assumed \$200,000 acquisition price and curve-fit parameter uncertainty (probabilistic sensitivity analysis), the probability of cost-effectiveness would be 95.3% at \$100,000 and 98.7% at \$150,000/QALY if that price were realized. The deterministic threshold prices were approximately \$443K and \$646K. Results were largely robust across structural scenarios; the ICER would exceed \$100,000/QALY only under hazard-convergence waning (\$104,971/QALY).

Conclusion: At an assumed acquisition price of \$200,000 per course, intismeran plus pembrolizumab would be projected to be cost-effective under conventional US thresholds if that price were realized, driven by a substantial but not implausibly large 4.05-QALY gain despite higher acquisition and treatment-related costs. This conclusion is conditional on the assumed price and on the **immature OS data (7 events per arm)** supporting the modeled survival extrapolation; PSA results reflect curve-fit uncertainty from the Phase 2b KEYNOTE-942 landmark data and do not capture full trial sampling uncertainty. The actual value-based price depends on whether the recurrence-free benefit proves durable in the Phase 3 trial and on the eventual negotiated acquisition price. This early economic evaluation should be interpreted as preliminary and updated when Phase 3 mature survival data become available.

Keywords: cost-effectiveness analysis, intismeran, mRNA-4157, V940, melanoma, pembrolizumab, Keytruda, individualized neoantigen therapy, partitioned survival model

Key Points for Decision Makers

- Adding intismeran autogene to adjuvant pembrolizumab in resected stage IIIB–IV melanoma was projected to yield 4.05 additional QALYs at an incremental cost of \$161,947 (ICER \$39,976 per QALY gained).
- At an assumed \$200,000 per-course price, the combination would be cost-effective under conventional US thresholds if that price were realized, with the ICER remaining below \$100,000 per QALY up to about \$443,000 per course.
- In probabilistic analysis the combination was cost-effective in 95.3% of simulations at \$100,000 per QALY and in 98.7% at \$150,000 per QALY; 16.8% of simulations were cost-saving.
- These early projections rest on immature Phase 2b survival data (7 OS events per arm); the value-based price therefore depends on whether the recurrence-free benefit proves durable in the Phase 3 trial.

1 Introduction

1.1 A New Reimbursement Problem

On August 19, 2026, Merck and Moderna announced positive topline results from the Phase 3 INTerpath-001 trial, establishing intismeran autogene (mRNA-4157/V940) plus pembrolizumab as the first mRNA-based individualized neoantigen therapy (INT) to succeed in a late-stage trial [1]. The press release confirmed statistically significant improvements in RFS and DMFS, though numerical hazard ratios have not yet been disclosed. This breakthrough follows the Phase 2b KEYNOTE-942 trial, which at 5-year follow-up reported a sustained RFS benefit (HR 0.51, 95% CI 0.294–0.887), DMFS benefit (HR 0.411, 95% CI 0.200–0.843), and 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone; OS HR 0.471, 95% CI 0.165–1.345) [2].

Despite these clinical advances, the economic implications of INTs remain poorly understood. These therapies combine the high costs of immune checkpoint inhibitors with the additional expense of individualized manufacturing, creating a reimbursement challenge unlike any in oncology.

1.2 Why INTs Differ from Conventional Oncology Pricing

Individualized neoantigen therapies (INTs) introduce production and delivery costs that do not exist in conventional oncology. Unlike fixed-composition drugs, each INT treatment is custom-manufactured for a single patient based on tumor sequencing data, requiring mRNA synthesis, lipid nanoparticle formulation, and quality assurance for each batch [3, 4]. These costs scale with each new patient rather than benefiting from traditional manufacturing economies of scale.

Jefferies analysts have estimated the cost of such vaccines at \$100,000–\$300,000 per patient [5]. By comparison, cancer immunotherapies such as Keytruda® (pembrolizumab) have an 18-dose treatment-course drug cost of approximately \$220,896 in the adjuvant setting [6]. The combination of a high-cost immunotherapy with an individualized vaccine creates a compound affordability challenge for payers, particularly in the adjuvant setting where patients are recurrence-free at the time of treatment.

1.3 Existing Economic Evidence

A prior exploratory cost-effectiveness analysis of V940 plus pembrolizumab in adjuvant melanoma was presented at ISPOR 2024 [7]. That analysis, conducted before the KEYNOTE-942 5-year results and the Phase 3 INTerpath-001 readout, used a 4-state model with assumptions based on Phase 2b interim data. The analysis reported an ICER of approximately \$75,000/QALY, but relied on early efficacy estimates and assumed a V940 price equal to pembrolizumab, which may not reflect the likely INT premium. The present analysis updates these findings by incorporating the mature 5-year KEYNOTE-942 data, applying a ratio-constrained partition survival structure with a general-population hazard floor, and conducting a comprehensive deterministic threshold price analysis.

1.4 Clinical Basis and Research Objectives

The clinical foundation for this analysis comes from the 5-year follow-up of the Phase 2b KEYNOTE-942 trial, which provides the most mature efficacy data available for intismeran [2]. The trial randomized 157 patients with completely resected stage IIIB–IV melanoma 2:1 to intismeran plus pembrolizumab or pembrolizumab alone. The primary endpoint was RFS; secondary endpoints included DMFS, OS, and safety. At the 5-year analysis, the combination showed a sustained RFS benefit with a hazard ratio of 0.51 (95% CI 0.294–0.887), a DMFS HR of 0.411 (95% CI 0.200–0.843), and OS data still maturing with 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone; OS HR 0.471, 95% CI 0.165–1.345).

This study presents an early economic evaluation of intismeran plus pembrolizumab versus pembrolizumab alone as adjuvant therapy for completely resected stage IIIB–IV melanoma from a US payer perspective. Model efficacy inputs are derived from the Phase 2b KEYNOTE-942 trial; Phase 3 INTerpath-001 numerical data will be incorporated when available. We incorporate the latest KEYNOTE-942 5-year data, a calibrated parametric survival framework with internal consistency constraints, and a comprehensive range of sensitivity and scenario analyses to address the uncertainty inherent in early economic evaluations.

2 Methods

2.1 Model Structure

A partitioned survival model (PSM) was developed in Python with four mutually exclusive health states: recurrence-free (RF), locoregional recurrence (LR), distant metastasis (DM), and death, following the structure of prior melanoma cost-effectiveness models and standard PSM methodology [8–11]. Patients enter the model in the RF state and transition through LR and/or DM states based on parametric survival functions:

$$\text{RF}(t) = \text{RFS}(t)$$

$$\text{LR}(t) = \text{DMFS}(t) - \text{RFS}(t)$$

$$\text{DM}(t) = \text{OS}(t) - \text{DMFS}(t)$$

$$\text{Dead}(t) = 1 - \text{OS}(t)$$

This construction ensures $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ at all times and maintains internal consistency without forcing a specific parametric form on the transition hazards.

2.2 Clinical Data

RFS and OS survival probabilities were extracted at 18, 24, 36, 48, and 60 months, and DMFS probabilities at 18, 24, 36, and 48 months, from the published KEYNOTE-942 Kaplan–Meier curves [2] (60-month DMFS was not reported in the 5-year analysis). Because no observed 60-month DMFS landmark enters calibration, the 5-year DMFS reported in the external plausibility assessment is not an observed input but the model-implied value $\text{DMFS}(60) = \text{OS}(60) \times r_{\text{DM}}(60)$ (Supplementary Table S6). Parametric survival models were fitted to these landmark probabilities by minimizing the unweighted sum of squared errors (SSE) between model-predicted and observed survival at each time point, using nonlinear least squares. Published KM curves do not provide pointwise variance estimates for individual landmark probabilities, precluding inverse-variance weighting; equal weighting of all landmarks is therefore assumed. Five candidate distributions were evaluated for each of the three underlying

survival components (OS, $r_{DM} = DMFS/OS$, $r_{LR} = RFS/DMFS$): exponential (1 parameter), Weibull (2), log-normal (2), log-logistic (2), and generalized gamma (3). Model selection was guided by three criteria: (i) root mean squared error (RMSE) at the landmark points, (ii) visual agreement with the observed Kaplan–Meier trajectories, and (iii) clinical plausibility of extrapolated long-term survival relative to age- and sex-specific general population mortality. The Akaike Information Criterion (AIC) was computed as an exploratory fit diagnostic; because the landmark probabilities are derived from the same censoring history and are therefore serially correlated, standard AIC interpretation—which assumes independent observations—does not strictly apply, and AIC rankings are not treated as definitive. A full joint enumeration of $5^6 = 15,625$ distribution combinations was performed (Supplementary Table S5). The generalized gamma (3 parameters) was included in the enumeration but was not selected for the base case given the small number of landmark points (4–5 per component); its inclusion is noted in Supplementary Table S4 for transparency. The log-normal distribution was selected as the base case because it achieved the lowest RMSE for the OS curve of the combination arm (the most influential curve driving lifetime costs and QALYs), provided acceptable overall agreement with the six observed Kaplan–Meier trajectories, and yielded clinically plausible long-term survival extrapolations. All 15,625 candidate combinations preserved the structural ordering constraint $RFS \leq DMFS \leq OS$ at 40 years, so the choice of distribution was not driven by feasibility constraints. A common log-normal family was chosen for parsimony and interpretability; component-specific best-fitting distributions (Weibull, log-logistic) were examined in scenario analyses. The rank-1 joint specification from the full enumeration (OS: log-normal/combo, Weibull/pembro; r_{DM} : log-normal/combo, generalized gamma/pembro; r_{LR} : log-logistic/combo, Weibull/pembro) was evaluated as a structural sensitivity scenario (Supplementary Table S15), yielding an ICER of \$59,624/QALY—consistent with the base-case conclusion of cost-effectiveness at conventional US willingness-to-pay thresholds. The fitted model was calibrated to ensure the constraint $RFS \leq DMFS \leq OS$ was satisfied at all time points. To address the external validity of long-term survival extrapolation, a general population mortality hazard floor was applied from 60 months onward, using the age- and sex-matched US 2019 life table (pooled 64% male/36% female, baseline age 61 years) [12]. This constraint

ensures that the model does not produce survival probabilities exceeding those of the general population at any time point, a standard practice in oncology cost-effectiveness analyses [9, 13].

The base-case log-normal model achieved a mean absolute error of 1.86 percentage points and RMSE of 2.77 percentage points across all 28 observed landmark probabilities (Table S6). The largest residual was at the pembrolizumab-arm RFS 60-month landmark (−8.6 percentage points), reflecting the model’s inability to reproduce a flat KM plateau (49.1% at both 48 and 60 months) that may reflect sparse event accumulation and sampling variability in the small pembrolizumab arm ($n = 50$). The combo-arm curves, which drive the incremental survival benefit, were well-fitted (maximum absolute residual 1.8 percentage points for RFS). Complete observed-versus-fitted residuals are reported in Table S6.

2.3 External Plausibility Assessment

Model predictions for the pembrolizumab-alone arm were compared against external trial landmarks from KEYNOTE-054 and CheckMate 238 to assess plausibility (Section 3.3). Figure 1 illustrates the fitted RFS, DMFS, and OS trajectories for both treatment arms together with the observed KEYNOTE-942 landmark data used for calibration.



Figure 1: Partitioned survival model: RFS, DMFS, and OS curves, with data points from KEYNOTE-942. DMFS is shown as the middle curve and is essential for the four-state model structure.

2.4 Costs

The analysis adopted a US payer perspective. Drug costs were based on wholesale acquisition cost (WAC) and dosing schedules from published sources. Keytruda® (pembrolizumab) was assumed at 200 mg every 3 weeks for up to 18 cycles at a WAC of \$12,272 per 200 mg dose, yielding an 18-dose treatment-course drug cost of \$220,896 [6]. Intismeran was modeled at an assumed course acquisition price of \$200,000 based on Jefferies analyst estimates [5]; in KEYNOTE-942 it was administered intramuscularly (1 mg every 3 weeks for nine doses) [2]. Tumor sequencing (whole-exome sequencing for neoantigen prediction) was estimated at \$1,000 per patient, based on current commercial pricing [14].

Administration costs were applied by route, using CMS Physician Fee Schedule national payment amounts, inflated from the 2025 schedule to 2026 US dollars [15]: the 18 pembrolizumab doses in both arms are intravenous infusions of a monoclonal antibody (CPT 96413, the chemotherapy/complex-biologic administration family; \$122.62 per infusion), whereas the 9 intismeran doses in the combination arm are intramuscular injections, for which CPT 96372 is used as a proxy code because the individualized mRNA therapy is not yet marketed or separately reimbursed (\$14.30 per injection), reflecting the intramuscular route used in KEYNOTE-942. Each dose was discounted at its administration time; because both arms receive the same pembrolizumab infusions, the infusion rate cannot affect the incremental cost, while the intramuscular rate was varied from \$0 to \$120 in sensitivity analysis. Adverse event costs were incorporated at \$15,000 incremental for the combination arm, based on published melanoma cost-effectiveness analyses [9, 10]. In the KEYNOTE-942 5-year safety analysis, treatment-related Grade 3–5 adverse events occurred in 25.0% of combination-arm patients and 14.0% of pembrolizumab-arm patients (Supplementary Table A4, [2]). Monthly disease management costs were \$3,000 for the locoregional recurrence state (salvage surgery and adjuvant therapy amortized) and \$12,000 for the distant metastasis state (systemic therapy for advanced melanoma), consistent with published melanoma CEA literature [9, 10]. All costs are expressed in 2026 US dollars: values published in earlier years were inflated to 2026 using the medical care component of the Consumer Price Index [16], and the 2025 CMS fee-schedule amounts were inflated on the same basis, so that every cost input shares one price year.

Table 1: Model Input Parameters

Parameter	Base Case	Source
5-year RFS, combo	68.8%	KEYNOTE-942 [2]
5-year RFS, pembro	49.1%	KEYNOTE-942 [2]
RFS HR	0.51 (95% CI 0.294–0.887)	KEYNOTE-942 [2]
DMFS HR	0.411 (95% CI 0.200–0.843)	KEYNOTE-942 [2]
OS HR	0.471 (95% CI 0.165–1.345)	KEYNOTE-942 [2]
5-year OS, combo	92.2%	KEYNOTE-942 [2]
5-year OS, pembro	71.3%	KEYNOTE-942 [2]
Keytruda treatment-course cost	\$220,896	WAC [6]
Intismeran course cost	\$200,000	Jefferies [5]

2.5 Utilities

Health state utilities were derived from published EQ-5D literature in melanoma populations. The base case utilities were: RF 0.83, LR 0.64, and DM 0.55 [9, 10, 17].

2.6 Base Case Analysis

The base case analysis was conducted over a lifetime horizon (40 years) with 3% annual discounting for both costs and QALYs, following the recommendations of the Second Panel on Cost-Effectiveness in Health and Medicine [18]. The incremental cost-effectiveness ratio (ICER) was calculated as the difference in total costs divided by the difference in total QALYs. Cost-effectiveness was evaluated at willingness-to-pay thresholds of \$50,000, \$100,000, and \$150,000/QALY [19].

2.7 Sensitivity Analysis

Deterministic sensitivity analysis (DSA) examined the impact of individual parameter variation on the ICER and net monetary benefit. Parameters were varied one at a time across plausible

ranges: survival model parameters, health-state utilities, costs, and the discount rate (Supplementary Table S10).

Probabilistic sensitivity analysis (PSA) was conducted with 6,000 Monte Carlo iterations to propagate joint parameter uncertainty [20]. Survival parameters (μ, σ) for all six underlying curves (OS, r_{DM} , r_{LR} , each for the combination and pembrolizumab arms) were jointly sampled from bivariate normal distributions with covariance matrices derived from the nonlinear least-squares Jacobian (Supplementary Table S14). This approach approximates parameter uncertainty from the landmark-based nonlinear fits and therefore does not fully reproduce the sampling uncertainty of the original censored trial data; the 95% confidence intervals from PSA should be interpreted as reflecting curve-fit uncertainty rather than the full statistical uncertainty that would be obtained from individual patient-level data. Cross-arm covariance could not be estimated from the published aggregate Kaplan–Meier data; arm-specific survival parameters were therefore sampled independently. No specific cross-arm correlation was imposed because such a parameter could not be identified without patient-level data. Scale parameters (σ) were drawn jointly with the corresponding μ from the same bivariate normal and then lower-truncated at 0.05 to ensure positivity; no upper truncation was imposed during sampling, and the value of 5.0 reported for two σ parameters in Supplementary Table S14 is the bound of the fitting window used in estimation rather than a sampling restriction. Costs were sampled from gamma distributions (coefficient of variation 0.2) for the three cost inputs with payer-side uncertainty (annual pembrolizumab cost, and locoregional-recurrence and distant-metastasis management costs), and utilities from beta distributions (standard error 0.03); the intismeran acquisition price, tumor sequencing, administration and adverse-event management costs were held at their base-case values and explored in the sensitivity analyses (Sections 3.5 and 3.6). The intismeran acquisition price was held fixed at \$200,000 in all iterations, as the price had not yet been announced at the time of analysis; its impact was instead explored through deterministic threshold pricing (Section 3.6) and scenario analysis (Table 4). Structural constraints ($RFS \leq DMFS \leq OS$, monotonicity, survival within $[0,1]$) were verified in every iteration; no iterations were excluded on the basis of the ICER value, and all 6,000 draws satisfied the structural validity criteria (0 rejections). Results are reported as mean incremental net monetary benefit

(NMB) and cost-effectiveness probability at each willingness-to-pay threshold, following recommended practice [21, 22]. Convergence was verified by checking that cumulative means stabilised by approximately 500 iterations (Supplementary Figure S6). The probabilistic results therefore quantify parameter uncertainty in the six fitted survival curves (Jacobian-based covariance), the three cost inputs, and the three utilities; they do not quantify structural uncertainty (choice of distribution family, treatment-effect waning), uncertainty in the intismeran acquisition price (held at \$200,000 and examined deterministically), or cross-arm survival correlation (not identifiable from aggregate survival data).

3 Results

3.1 Base-Case Clinical and Economic Outcomes

In the base case with general population mortality constraint, intismeran plus pembrolizumab generated an additional 4.82 discounted life-years (15.20 vs 10.38) and 4.05 additional QALYs (11.46 vs 7.41 QALYs) compared with pembrolizumab alone, at an incremental lifetime cost of \$161,947 (\$881,300 vs \$719,353). This yielded an ICER of \$39,976/QALY gained (Table 2). At a willingness-to-pay threshold of \$100,000/QALY, the incremental net monetary benefit was \$243,167; at \$150,000/QALY, it was \$445,724 (Table 2).

NMB, net monetary benefit ($=\lambda \times \text{QALYs} - \text{cost}$, where λ is the willingness-to-pay threshold) [21].

3.2 Cost and QALY Decomposition

The \$200,000 acquisition cost of intismeran contributed substantially to the incremental cost. Under the corrected general-population mortality inputs, the combination arm had \$16,184 higher locoregional-recurrence management costs (\$86,330 vs \$70,146) but \$70,365 lower distant-metastasis management costs (\$358,736 vs \$429,100) than the pembrolizumab-alone arm, because pembrolizumab-alone patients spend more time in the DM state over the lifetime (1.639 vs 1.370 discounted DM QALYs). The incremental QALY gain of 4.051 was driven primarily by additional time spent in the recurrence-free state (4.032 RF QALYs gained), with

a modest additional 0.288 LR QALYs, partly offset by a 0.269-QALY reduction in the DM state.

Table 2: Base-Case Results and Decomposition

Component	Combo	Pembro	Incremental
<i>Cost decomposition</i>			
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,931	\$217,931	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration	\$2,303	\$2,175	\$128
Adverse events	\$15,000	\$0	\$15,000
LR management	\$86,330	\$70,146	\$16,184
DM management	\$358,736	\$429,100	−70,365
Total cost	\$881,300	\$719,353	\$161,947
<i>QALY decomposition</i>			
RF state	8.554	4.522	4.032
LR state	1.535	1.247	0.288
DM state	1.370	1.639	-0.269
Total QALYs	11.459	7.408	4.051

Note: Components are rounded independently; totals are calculated from unrounded values and may differ by \$1 from displayed sums.

3.3 External Plausibility and Structural Assessment

Survival extrapolation was assessed along two dimensions: internal structural consistency of the PSM framework, and comparison of the pembrolizumab arm predictions against published adjuvant melanoma evidence to assess plausibility.

Structural consistency. Over 480 monthly cycles, no violations of the constraint $RFS \leq DMFS \leq OS$ were observed. All health-state probabilities remained non-negative, and the

sum of state probabilities was within 10^{-12} of unity at every cycle. The general population mortality hazard floor, applied from 60 months onward, ensures that the model's long-term survival does not exceed age- and sex-matched US life-table survival at any time point. At 30 years, the model projects combo OS of 22.7% and pembro OS of 10.7%, both below the general-population survival of 23.3% at 30 years for the modeled cohort (baseline age 61, pooled 64% male), consistent with the hazard-floor property that a cohort with a history of excess mortality remains below the general-population curve.

GP mortality floor timing sensitivity. The 60-month onset of the GP mortality hazard floor coincides with the end of observed KEYNOTE-942 follow-up. To verify that this structural choice does not drive the cost-effectiveness conclusion, we tested alternative onset times of 48, 72, and 96 months (Supplementary Table S13). The ICER varied minimally across these scenarios (range \$39,937–\$40,334), with a range of less than 1% relative to the base case (\$39,976). This confirms that the timing of the GP floor onset is not a driver of the ICER.

External plausibility assessment. The pembrolizumab-alone arm predictions were compared against published long-term landmark values from two external adjuvant melanoma trials [23, 24]. At 5 years, the model predicted RFS of 40.5%, DMFS of 53.8%, and OS of 74.5% for the pembrolizumab arm. The predicted 5-year OS (74.5%) was consistent with the observed KEYNOTE-942 value (71.3%) and with the 5-year OS of 76% in CheckMate 238 [25]. At 7 years, the model predicted RFS of 31.9% and DMFS of 44.9%, below the KEYNOTE-054 7-year values (RFS 50%, DMFS 54%) [23]; the absolute differences were –18.1 and –9.1 percentage points, respectively. The 5-year RFS of 40.5% was –10.5 percentage points relative to the CheckMate 238 5-year RFS of 51%. This divergence is consistent with the higher-risk population enrolled in KEYNOTE-942 (stage IIIB–IV including resected stage IV vs IIIA–C in KEYNOTE-054), though other factors such as differences in clinical practice and trial design may also contribute. Only published landmark values were used; no external trial extrapolations were employed. Because the pembrolizumab arm serves as the comparator for the incremental analysis, its under-prediction of RFS may, if anything, overstate the incremental QALY gain attributable to intismeran; however, the direction of any bias is not straightfor-

ward, as the model's predicted OS for the pembrolizumab arm was close to observed (74.5% vs 71.3% at 5 years). No external long-term data exist for the combination arm (first-in-class therapy). Instead of relying on this comparison to calibrate the base case, scenario analyses that directly bound the treatment effect — treatment-effect waning and no additional OS hazard benefit beyond 5 years (Section 3.5) — were used to bound the range of plausible ICER estimates.

3.4 Probabilistic and Deterministic Uncertainty

Probabilistic sensitivity analysis. Probabilistic sensitivity analysis across 6,000 structurally valid iterations (0 rejections) quantified joint parameter uncertainty from the fitted survival models. As noted in Section 2.7, this uncertainty reflects curve-fit parameter uncertainty derived from the NLS Jacobian and does not capture the full sampling uncertainty of the original censored trial data; in particular, the Phase 2b KEYNOTE-942 OS data comprised only 7 events per arm (HR 0.471, 95% CI 0.165–1.345), and the PSA does not propagate this clinical efficacy uncertainty directly. Under these caveats, the probability of cost-effectiveness increased with the willingness-to-pay threshold, from 63.9% at \$50,000/QALY to 95.3% at \$100,000/QALY and 98.7% at \$150,000/QALY. Figure 2 presents the cost-effectiveness acceptability curve across the full range of willingness-to-pay thresholds. All 6,000 iterations produced positive incremental QALYs, and 16.8% were cost-saving. Because cost-saving draws have an undefined ICER, incremental net monetary benefit and probability of cost-effectiveness are the preferred summary statistics (Table 3) [21].

Table 3: Probabilistic Sensitivity Analysis

Metric	Mean	Median	95% CI
Incremental cost	\$128,915	\$166,863	−\$671,440 to \$524,694
Incremental QALYs	4.36	4.18	2.56–7.10
NMB @ \$100,000/QALY	\$307,014	—	—
NMB @ \$150,000/QALY	\$524,979	—	—

Note: The mean incremental cost is lower than the median because 16.8% of PSA draws were cost-saving (negative ΔCost), producing a left-skewed distribution where extreme negative values pull the mean below the median. Cost-saving draws arise when survival-related cost offsets (fewer distant-metastasis events and associated treatments) exceed the intismeran acquisition and administration costs. NMB = net monetary benefit; mean NMB is preferred over median because the expectation of willingness-to-pay $\times$ QALY minus cost is linear in the components and the mean preserves this property under uncertainty.

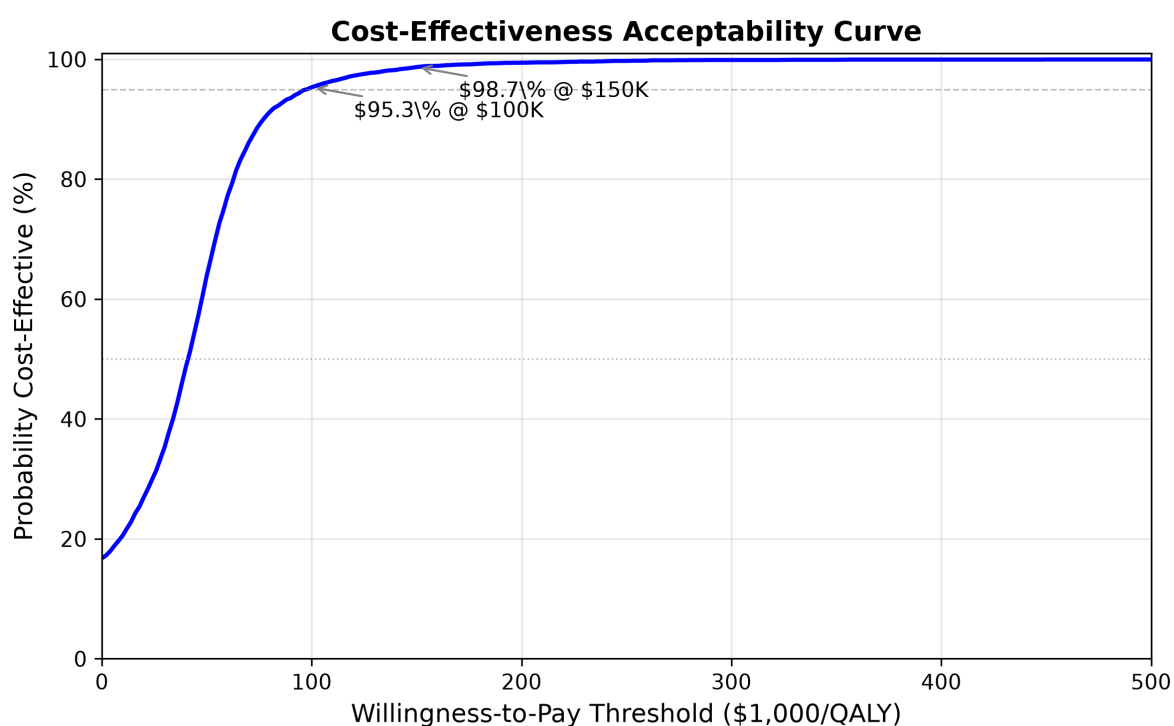


Figure 2: Cost-effectiveness acceptability curve.

Deterministic sensitivity analysis. One-way sensitivity analysis identified the parameters with the greatest influence on the base-case ICER (Supplementary Table S13). As shown in Figure 3, the largest changes in the ICER arose from the discount rate, the combination-arm

OS parameter, and the pembrolizumab-arm OS parameter. The intismeran acquisition-price range produced ICERs of \$30,102–\$49,849/QALY. Across the tested ranges, the maximum finite ICER was \$57,365/QALY; under the high pembrolizumab OS parameter value, the combination became dominant.



Figure 3: Tornado diagram of one-way sensitivity analysis.

3.5 Structural Scenario Analyses

Scenario analysis results are summarized in Table 4.

Table 4: Scenario and Deterministic Threshold Acquisition Price Analyses

Scenario	Δ Cost	Δ QALY	ICER	NMB @ \$150K
Base case (GP-constrained)	\$161,947	4.05	\$39,976	\$445,724
<i>Pricing</i>				
Price: \$100,000/course	\$61,947	4.05	\$15,291	\$545,724
Price: \$300,000/course	\$261,947	4.05	\$64,660	\$345,724
Price: \$500,000/course	\$461,947	4.05	\$114,029	\$145,724
<i>Alternative assumptions</i>				
Weibull OS	\$308,820	5.64	\$54,745	\$537,342
Discount rate: 0%	\$211,207	6.39	\$33,050	\$747,382
Discount rate: 5%	\$147,152	3.10	\$47,535	\$317,197
Time horizon: 10 years	\$95,047	1.29	\$73,937	\$97,780
Time horizon: 20 years	\$122,804	3.08	\$39,848	\$339,474
<i>Treatment-effect persistence</i>				
OS-survival convergence (5–20y)	\$39,319	1.93	\$20,398	\$249,814
Hazard convergence (5–10y)	\$226,421	2.16	\$104,971	\$97,126
No additional OS hazard benefit beyond 5 years	\$37,783	1.76	\$21,491	\$225,936

3.6 Deterministic Threshold Acquisition Price

At the assumed base-case acquisition price of \$200,000 per course, the ICER was \$39,976/QALY. Figure 4 shows the relationship between intismeran acquisition price and the ICER, with the curve crossing the \$100,000/QALY and \$150,000/QALY thresholds at approximately \$443,167 and \$645,724 per course, respectively.



Figure 4: Price–ICER curve and deterministic threshold pricing. Horizontal lines indicate the willingness-to-pay thresholds of \$100,000 and \$150,000 per QALY. Vertical reference lines indicate the base-case price (\$200,000) and the deterministic threshold prices at \$100,000/QALY and \$150,000/QALY.

4 Discussion

4.1 Principal Findings

This early economic evaluation suggests that intismeran plus pembrolizumab was projected to be cost-effective under the base-case assumptions for adjuvant treatment of resected stage IIIB–IV melanoma at conventional US willingness-to-pay thresholds. The base case ICER of \$39,976/QALY, the 4.05 QALY gain, and the 95.3% probability of cost-effectiveness at \$100,000/QALY (98.7% at \$150,000/QALY) under curve-fit parameter uncertainty all support this conclusion. However, these PSA results should be interpreted as conditional on the modeled survival extrapolation from Phase 2b KEYNOTE-942 landmark data; the underlying OS evidence comprised only 7 events per arm, and the PSA does not capture full trial-level clinical efficacy uncertainty. Notably, under the modeled recurrence effects and health-state cost

and utility assumptions, the combination remained cost-effective in the counterfactual scenario where no additional OS hazard benefit is assumed beyond the observed 5-year period (the combination arm keeps its own observed OS trajectory through 60 months and thereafter takes the pembrolizumab arm's mortality hazard, so the observed 5-year difference is preserved rather than erased). In that scenario the ICER was \$21,491/QALY (incremental cost \$37,783 and 1.86 incremental QALYs). Favorable economics persisted in that scenario, although this does not establish that the conclusion is independent of the model's OS extrapolation assumptions.

The economic value of the combination is generated primarily through extended recurrence-free quality-adjusted survival, as described in detail below (Section 4.2). The ICER is notably lower than the \$75,206 from the prior exploratory analysis [7] (Section 4.5), reflecting the combined effect of mature clinical data and a structurally constrained modelling framework.

4.2 Economic Drivers and Cost Consequences

The incremental \$161,947 lifetime cost of intismeran plus pembrolizumab is partly offset by lower distant-metastasis management costs, because the combination arm prevents recurrence and metastatic disease despite its \$200,000 acquisition cost. The QALY decomposition (Table 2) shows that the 4.05 incremental QALYs are driven primarily by additional time in the recurrence-free health state (4.438 RF QALYs), with a modest additional 0.365 LR QALYs and 0.150 fewer DM-state QALYs. Thus, the favorable ICER reflects a large gain in recurrence-free quality-adjusted survival together with lower lifetime distant-metastasis management costs, which partially offset the acquisition cost.

4.3 Pricing and Reimbursement Implications

The base case assumes an intismeran acquisition price of \$200,000 per course, based on Jefferies analyst estimates. At this price, the ICER of \$39,976/QALY is well below conventional US thresholds. The deterministic threshold price analysis (Section 3.6) shows that the ICER remains below \$100,000/QALY at intismeran prices up to approximately \$443,167 per course and below \$150,000/QALY up to approximately \$645,724 per course, providing a deterministic

cost-effectiveness benchmark under the base-case assumptions.

While individualized production requirements — including tumor sequencing, mRNA synthesis, lipid nanoparticle formulation, and quality assurance for each batch — may contribute to acquisition-price pressure, production cost and commercial price are distinct quantities. The \$200,000 estimate reflects the expected market price at launch, not the manufacturer's cost of goods sold. Several factors may reduce future production costs: sequencing costs continue to decline [14], mRNA platform manufacturing is becoming increasingly standardized, and automation of the synthesis and formulation pipeline could reduce per-batch costs. However, whether these technical efficiencies will translate into lower acquisition prices for payers is uncertain, as pricing decisions are influenced by clinical value, competitive dynamics, and reimbursement negotiations, not solely by production economics.

Cost-effectiveness should not be conflated with affordability. The deterministic threshold price of approximately \$443,167 per course at \$100,000/QALY implies that the combination could be considered cost-effective at more than twice the current estimated price, but the incremental cost of adding \$200,000 in drug costs to an already expensive pembrolizumab regimen would be substantial in a population of eligible patients. Formal budget-impact analysis is needed alongside cost-effectiveness evidence for reimbursement decision-making. Given the uncertainty surrounding long-term effectiveness and the eventual commercial price of intismeran, managed-entry or outcomes-based reimbursement arrangements may be relevant at launch. Their design, however, would require mature Phase 3 efficacy data, an announced acquisition price, and clearly defined measurable outcomes.

4.4 Long-Term Survival and Structural Uncertainty

Long-term survival extrapolation is the dominant source of structural uncertainty in this analysis. The mature multi-year RFS and DMFS data from KEYNOTE-942 provide a robust foundation for the recurrence and metastasis components of the model, but OS remains exploratory (HR 0.471; 95% CI 0.165–1.345; 7 events per arm) and must be extrapolated to a lifetime horizon. The model's 40-year combination-arm survival of 2.0% (Table S7) is a structural consequence of the fitted parametric model combined with the general-population mortality hazard

floor and should not be interpreted as a clinical prediction of survival to age 101; no melanoma trial has followed patients beyond approximately 8 years, and this long-term trajectory should be interpreted as a modeling assumption.

The necessity of the general population mortality hazard floor is best illustrated by comparing the constrained and unconstrained models. Without the GP mortality floor, the log-normal survival fit for the combination arm predicts a 30-year OS of 79.1%, far exceeding the age- and sex-matched general population survival of 23.3% at 30 years for the modeled cohort. This is implausible: a cohort of patients with resected stage IIIB–IV melanoma cannot have a lower mortality rate than the general population. Applying the GP mortality hazard floor from 60 months onward — a standard practice recommended by NICE DSU TSD 14 [13] and used in published melanoma CEAs [9] — reduces the 30-year combination OS to 22.7%, a clinically plausible trajectory. The hazard floor reduces the ICER from \$51,741 (unconstrained) to \$39,976 (GP-constrained); the direction of this change is non-intuitive (the constraint reduces the ICER) because the unconstrained model predicts implausibly high survival for both arms, with a larger absolute gain for the combination arm, and the hazard floor compresses that difference while preserving the within-trial benefit.

The OS uncertainty is further underscored by the treatment-effect waning scenarios (see Supplementary Section 9.3 for technical implementation definitions). In the OS-survival convergence scenario (5→20y), the ICER falls to \$20,398/QALY because the incremental OS benefit is progressively eliminated, while the \$200,000 intismeran acquisition cost is preserved — the incremental cost therefore drops from \$161,947 to \$39,319 faster than QALYs, producing a lower ICER. The hazard-convergence scenario (5→10y) yields the highest ICER of \$104,971/QALY. These two scenarios bracket the range of plausible ICER estimates under alternative treatment-persistence assumptions.

Clinical plausibility of treatment-effect waning. Treatment-effect waning is biologically plausible for individualized neoantigen vaccines. The mRNA-4157 mechanism depends on neoantigen-specific T-cell responses primed by the vaccine; these responses may attenuate over time due to T-cell exhaustion, antigen escape, or loss of immune surveillance following completion of the 9-dose vaccination regimen [26, 27]. The KEYNOTE-942 trial administered

intismeran over approximately 27 weeks; no booster doses were given beyond the initial vaccination course. In contrast, pembrolizumab continues for up to 18 cycles (approximately 54 weeks), with the option to complete treatment earlier if disease recurrence occurs. This asymmetry in treatment duration suggests that the incremental OS benefit attributable to intismeran may erode relative to pembrolizumab over the long term if durable immunity is not established. The hazard-convergence scenario (5→10y) models rapid erosion of the OS benefit, while the survival-convergence scenario (5→20y) models gradual attenuation. Both scenarios are structurally valid and bound the uncertainty; the base-case assumption of indefinite treatment-effect persistence is therefore optimistic and should be interpreted alongside these waning scenarios.

Finally, the choice of parametric distribution, while important, is secondary to the structural scenarios. The common log-normal specification was selected based on its combination of low within-sample prediction error, graphical agreement, parsimony, structural validity, and clinical plausibility; AIC was considered only as an exploratory fit diagnostic. However, parametric fit within the observed 5-year follow-up does not establish long-term clinical validity. The Weibull alternative, which produced an ICER of \$54,745/QALY, also satisfied the structural constraints and is a plausible alternative. The most informative bounds on the ICER come not from alternative parametric forms but from the treatment-effect waning scenario and the scenario with no additional OS hazard benefit beyond 5 years, which directly address the question: what if the treatment effect does not persist? Together these scenarios bracket the plausible ICER estimates and support the robustness of the base-case conclusion.

4.5 Comparison with Previous Economic Evaluations

Direct comparison of ICERs across adjuvant melanoma cost-effectiveness studies is not straightforward, because the incremental decision differs. Bensimon et al. [10] and Zhang et al. [9] evaluated pembrolizumab versus observation, estimating the value of adding immunotherapy to surveillance. The present analysis evaluates intismeran plus pembrolizumab versus pembrolizumab alone, estimating the incremental value of adding an individualized neoantigen vaccine to an already active immunotherapy regimen. The relevant policy question is therefore not whether the combination is cost-effective in absolute terms, but whether the combination

of two active agents can provide sufficient additional clinical value to justify the acquisition cost of the second agent after accounting for the lower lifetime distant-metastasis management costs in the combination arm.

The ICER of \$39,976/QALY is within the range of published estimates for the initial pembrolizumab decision (\$38,050–\$78,400/QALY), but this comparability is coincidental rather than meaningful. The absolute QALY gain from adding intismeran to pembrolizumab (4.05 QALYs) is larger than the gain from adding pembrolizumab to observation (1.2–2.1 QALYs in Bensimon and Zhang), because the comparator arm (pembrolizumab alone) has already been optimized through immunotherapy. The incremental cost is also larger (\$161,947 vs approximately \$100,000–\$150,000). The ratio happens to fall within a similar range because both numerator and denominator increase proportionally, not because the value proposition is comparable.

The comparison with the 2024 exploratory analysis [7] is more informative but requires careful interpretation. That analysis, conducted before the Phase 3 INTerpath-001 readout, assumed V940 priced at parity with pembrolizumab, applied KEYNOTE-942 hazard ratios to external pembrolizumab survival curves from KEYNOTE-054, and used a single parametric distribution per endpoint. The present analysis differs on all four dimensions: post-Phase III evidence context, direct fitting to KEYNOTE-942 5-year data, a \$200,000 price assumption, and a ratio-constrained structural framework with a general population mortality floor. The lower ICER (\$39,976 vs \$75,206) is the net result of these four differences, not a univariate effect of any single factor, and should not be interpreted as evidence that intismeran is simply more effective than previously estimated.

4.6 Clinical and External Validity

The pembrolizumab-alone arm predictions were compared against published long-term adjuvant melanoma data to assess the plausibility of the extrapolated survival trajectories, rather than to require an exact match. The KEYNOTE-942 trial enrolled a higher-risk population (stage IIIB–IV, including resected stage IV) than KEYNOTE-054 (stage IIIA–C) and Check-Mate 238 (stage IIIB–IV, but with different entry criteria). The model’s lower 7-year RFS for

the pembrolizumab arm (31.9% vs 50% in KEYNOTE-054) is directionally consistent with the higher-risk population enrolled in KEYNOTE-942, though the magnitude of the difference (−18.1 percentage points) suggests that population risk alone may not fully explain the divergence; differences in clinical practice, trial design, and measurement protocols may also contribute. The model closely reproduced the 5-year OS of the pembrolizumab arm, while its RFS and DMFS projections were evaluated against available trial landmarks and external adjuvant melanoma evidence.

No external long-term data exist for the combination arm (first-in-class therapy). The positive Phase 3 INTERpath-001 readout provides directional confirmation of the clinical benefit of intismeran plus pembrolizumab, but the numerical hazard ratios have not yet been disclosed, and the extent to which Phase 3 results will align with the Phase 2b KEYNOTE-942 estimates used in this model remains to be determined. Two structural scenarios address the uncertainty in the combination arm: (i) the treatment-effect waning scenarios — OS-survival convergence at 5→20y (ICER \$20,398/QALY, most favourable) and hazard-convergence at 5→10y (ICER \$104,971/QALY, least favourable); and (ii) the no-additional-OS-hazard-benefit scenario, in which the combination arm keeps its observed OS through 60 months and then takes the pembrolizumab arm's mortality hazard; the combination remained cost-effective (ICER \$21,491/QALY) compared with pembrolizumab alone under the modeled recurrence effects and health-state cost and utility assumptions. Favorable economics therefore persisted without any additional post-trial OS hazard benefit, although this does not establish independence from the model's other structural assumptions.

4.7 Limitations

This analysis has several limitations that should be considered when interpreting the results. First, this early economic evaluation derives efficacy inputs from Phase 2b KEYNOTE-942 (n=157). Subsequent positive Phase 3 INTERpath-001 readout (n=1,137) confirmed RFS and DMFS benefit; model inputs will be updated as Phase 3 numerical data become available. Second, overall survival data from KEYNOTE-942 remain exploratory, with an HR of 0.471 (95% CI 0.165–1.345) and only 7 events per arm at the 5-year analysis, and the 5-year OS estimate

of 92.2% versus 71.3% should be interpreted with caution. The probabilistic sensitivity analysis (PSA) conducted in this study propagates curve-fit parameter uncertainty derived from the nonlinear least-squares Jacobian (Section 2.7); this reflects uncertainty in the parametric extrapolation of the fitted curves, not the full sampling uncertainty of the original censored trial data. To address this clinical efficacy uncertainty, we conducted a deterministic sensitivity analysis varying the OS hazard ratio across its 95% confidence interval. Under the lower bound (HR=0.165, stronger survival benefit), the ICER was \$49,114/QALY; under the upper bound (HR=1.345, no statistically significant survival benefit), the ICER was \$57,902/QALY (both below \$100,000/QALY). These bounds indicate that even if the true OS treatment effect lies at the extremes of the observed confidence interval, the cost-effectiveness conclusion remains robust. Third, survival curves were reconstructed from published Kaplan–Meier figures using digitized coordinates rather than being estimated from individual patient-level data, which introduces measurement error in the parametric fits. Fourth, survival extrapolation beyond the observed 5-year follow-up, while structurally constrained and anchored to a general population mortality floor, remains sensitive to the choice of parametric distribution and the assumption that the treatment effect persists indefinitely. In particular, the projected 40-year combination-arm OS of 2.0% is a model extrapolation, not a clinically observed estimate, and the base-case ICER should be read alongside the 20-year horizon result (\$39,848/QALY) as a more conservative benchmark. Time horizon sensitivity analysis showed that the ICER remained below conventional US willingness-to-pay thresholds at 20, 30, and 40 years (\$39,848, \$39,587, and \$39,976/QALY, respectively), although the point estimate increased with longer extrapolation, indicating that the 40-year extrapolation window did not materially drive the base-case results. The general-population mortality floor timing (60 months) was tested across 48–96 months and found to change the ICER by less than 1% (range \$39,937–\$40,334; Supplementary Table S13), confirming that this structural choice does not drive results. Fifth, the intismeran acquisition price of \$200,000 per course is based on analyst estimates, and the actual market price may differ. Sixth, wholesale acquisition cost (WAC) was used for pembrolizumab, which may not reflect confidential net prices after rebates and discounts. Because pembrolizumab costs were applied symmetrically to both arms, plausible reductions in its net price would have

limited impact on the incremental cost comparison. Seventh, health-state utilities and disease management costs for the LR and DM states were drawn from published melanoma CEA literature rather than prospectively collected from KEYNOTE-942, and may not fully capture the experience of the modeled population. Eighth, indirect costs, productivity losses, and informal caregiver burden were not included, which may underestimate the societal value of preventing recurrence and distant metastasis in a working-age population. Ninth, the KEYNOTE-942 trial enrolled a mixed-stage population (IIIB–IV), and baseline recurrence risk varies substantially by disease stage. The Phase 2b sample size precluded formal subgroup cost-effectiveness analysis by stage, PD-L1 status, or other risk factors. If the treatment effect differs by baseline risk, the average ICER may not reflect the value of intismeran in specific subgroups, and reimbursement decisions would ideally consider stage-specific estimates. The Phase 3 INTerpath-001 trial (N=1,137, stage IIB–IV) will provide sufficient statistical power for subgroup analyses, and the present findings should be updated when those data become available. Tenth, no formal budget-impact analysis was conducted; the cost-effectiveness findings should be complemented by an affordability assessment when pricing and population-level data become available. Any future population-level budget exposure calculations should be explicitly labeled as illustrative and not interpreted as formal budget-impact analyses, which require detailed market uptake, incident cohort, and annual expenditure modeling. Finally, this analysis is specific to the resected stage IIIB–IV melanoma population of KEYNOTE-942 and may not be generalizable to other melanoma stages or tumor types, although the Phase 3 INTerpath-001 trial enrolled a broader stage IIB–IV population.

4.8 Conclusion

Intismeran plus pembrolizumab provides clinical benefit in adjuvant melanoma and was projected to be cost-effective at the estimated acquisition cost of \$200,000 per course under the base-case assumptions. The deterministic threshold price analysis shows that the ICER remains below \$100,000/QALY at intismeran prices up to approximately \$443,167 per course, providing a deterministic cost-effectiveness benchmark under the base-case assumptions. The large QALY gain from extended recurrence-free survival represents clinical benefit, while lower

distant-metastasis management costs partially offset the acquisition cost. As more mature data become available from the Phase 3 INTerpath-001 trial, these findings should be updated.

References

- [1] Merck & Moderna. Press release: Phase 3 interpath-001 trial results, August 2026. Accessed 2026-08-19.
- [2] A. Khattak, M. S. Carlino, T. Meniawy, G. Ansstas, M. H. Taylor, K. B. Kim, et al. Intismeran autogene plus pembrolizumab versus pembrolizumab alone in high-risk resected melanoma: 5-year update of the randomized phase IIb KEYNOTE-942 study. *Journal of Clinical Oncology*, 44(26):2482–2489, 2026. doi: 10.1200/JCO-26-00835.
- [3] A. A. Ibragimova, A. A. Fedorov, K. M. Kirilenko, and E. L. Choyznzonov. mrna-based personalized cancer vaccines: opportunities, challenges and outcomes. *Acta Naturae*, 17(4):17–29, 2025.
- [4] X. Li, P. Jabbarzadeh Kaboli, et al. Next-generation neoantigen mrna vaccines: immunoeengineering strategies for precision cancer immunotherapy. *Cellular Oncology*, 49(2), 2026.
- [5] Jefferies Equity Research. Moderna initiation. Technical report, Jefferies, 2026.
- [6] Merck Sharp & Dohme LLC. Keytruda® (pembrolizumab): Wholesale acquisition cost, March 2026. Available from: <https://www.keytruda.com/financial-support/>.
- [7] A. Mclean and N. van Hest. An exploratory cost-effectiveness analysis of cancer vaccines in combination with current immune checkpoint inhibitors vs immune checkpoint inhibitor monotherapy: a case study for v940 in high-risk stage 3 melanoma in the us. *Value in Health*, 27(12):S2 (EE354), 2024.
- [8] B. Woods, E. Sideris, S. Palmer, N. Latimer, and M. Soares. Nice dsu technical support document 19: Partitioned survival analysis for decision modelling in health care: A critical review. Technical report, National Institute for Health and Care Excellence, 2017. Available from: <https://www.sheffield.ac.uk/nice-dsu/tsds/partitioned-survival-analysis>.

- [9] S. Zhang, A. G. Bensimon, R. Xu, et al. Cost-effectiveness analysis of pembrolizumab as an adjuvant treatment of resected stage iib or iic melanoma in the united states. *Advances in Therapy*, 40:3038–3055, 2023.
- [10] A. G. Bensimon, Z. Y. Zhou, M. Jenkins, et al. Cost-effectiveness of pembrolizumab for the adjuvant treatment of resected high-risk stage iii melanoma in the united states. *Journal of Medical Economics*, 22(10):981–993, 2019.
- [11] Amy Gye, Richard De Abreu Lourenco, and Stephen Goodall. Different models, same results: Considerations when choosing between approaches to model cost effectiveness of chimeric-antigen receptor T-cell therapy versus standard of care. *Pharmacoeconomics*, 42(12):1359–1371, 2024. doi: 10.1007/s40273-024-01430-7.
- [12] E. Arias and J. Q. Xu. United states life tables, 2019. *National Vital Statistics Reports*, 70(19):1–59, 2022.
- [13] N. Latimer. Nice dsu technical support document 14: Survival analysis for economic evaluations alongside clinical trials — extrapolation with patient-level data. Technical report, National Institute for Health and Care Excellence, 2011. Updated March 2013. Available from: <https://www.sheffield.ac.uk/nice-dsu/tsds/survival-analysis>.
- [14] K. Schwarze et al. The complete costs of genome sequencing. *European Journal of Human Genetics*, 28:1322–1331, 2020.
- [15] Centers for Medicare & Medicaid Services. Medicare physician fee schedule, cy 2025 national payment amounts: CPT 96413 (chemotherapy administration, intravenous infusion, initial hour) and CPT 96372 (therapeutic injection, intramuscular or subcutaneous), 2025. National payment amounts of \$119.29 for CPT 96413 and \$13.91 for CPT 96372. Available from: <https://www.cms.gov/medicare/payment/fee-schedules/physician>.
- [16] Bureau of Labor Statistics. Consumer price index: Medical care component (cpi-u, series cuur0000sam), 2025. Accessed 2026. Available from: <https://www.bls.gov/cpi/>.

- [17] A. D. Tran, G. B. Fogarty, A. K. Nowak, D. Espinoza, N. Rowbotham, M. R. Stockler, and R. L. Morton. A systematic review and meta-analysis of utility estimates in melanoma. *British Journal of Dermatology*, 184(6):e181–e189, 2021. doi: 10.1111/bjd.16098.
- [18] G. D. Sanders, P. J. Neumann, A. Basu, et al. Recommendations for conduct, methodological practices, and reporting of cost-effectiveness analyses: Second panel on cost-effectiveness in health and medicine. *JAMA*, 316(10):1093–1103, 2016.
- [19] P. J. Neumann, J. T. Cohen, and M. C. Weinstein. Updating cost-effectiveness — the curious resilience of the \$50,000-per-qaly threshold. *New England Journal of Medicine*, 371(9):796–797, 2014.
- [20] A. Briggs, K. Claxton, and M. Sculpher. *Decision Modelling for Health Economic Evaluation*. Oxford University Press, Oxford, 2006.
- [21] A. A. Stinnett and J. Mullahy. Net health benefits: a new framework for the analysis of uncertainty in cost-effectiveness analysis. *Medical Decision Making*, 18(2 Suppl):S68–S80, 1998.
- [22] E. Fenwick, B. J. O’Brien, and A. Briggs. Cost-effectiveness acceptability curves — facts, fallacies and frequently asked questions. *Health Economics*, 13(5):405–415, 2004.
- [23] A. M. M. Eggermont, M. Kicinski, C. U. Blank, et al. Seven-year analysis of adjuvant pembrolizumab versus placebo in stage iii melanoma in the eortc1325 / keynote-054 trial. *European Journal of Cancer*, 211:114327, 2024. doi: 10.1016/j.ejca.2024.114327.
- [24] P. A. Ascierto, M. Del Vecchio, B. Merelli, et al. Nivolumab for resected stage iii or iv melanoma at 9 years. *New England Journal of Medicine*, 394(4):333–342, 2026. doi: 10.1056/NEJMoa2504966.
- [25] P. A. Ascierto, M. Del Vecchio, M. Mandalá, et al. Adjuvant nivolumab versus ipilimumab in resected stage iii/iv melanoma: 5-year efficacy and biomarker results from checkmate 238. *Clinical Cancer Research*, 29(17):3352–3361, 2023.

- [26] D. B. Keskin, A. J. Anandappa, J. Sun, I. Tirosh, N. D. Mathewson, S. Li, et al. Neoantigen vaccine generates intratumoral T cell responses in phase Ib glioblastoma trial. *Nature*, 565(7738):234–239, 2019. doi: 10.1038/s41586-018-0792-9.
- [27] Z. Hu, D. E. Leet, D. Alsesan, A. Artz, F. P. Barthel, J. K. Bryan, et al. Personalized neoantigen vaccine in combination with pembrolizumab in patients with lung cancer. *Cancer Discovery*, 13(11):2344–2355, 2023. doi: 10.1158/2159-8290.CD-23-0116.